

Extra Unit B practice

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity B.

Moments and centroid

$$M_x = \frac{\rho}{2} \int_a^b (f(x)^2 - g(x)^2) dx$$

$$M_y = \rho \int_a^b x(f(x) - g(x)) dx$$

$$m = \rho \int_a^b (f(x) - g(x)) dx$$

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right)$$

Area between curves

$$A = \int_a^b (f(x) - g(x)) dx$$

Work

$$W = \int_a^b F(x) dx$$

Hydrostatic force

$$F = \rho g \int_a^b w(y) d(y) dy$$

Surface area

$$A = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

$$A = \int_a^b 2\pi x \sqrt{1 + f'(x)^2} dx$$

Arc length

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx$$

Volumes

Cross sections: $V = \int_a^b A(x) dx$

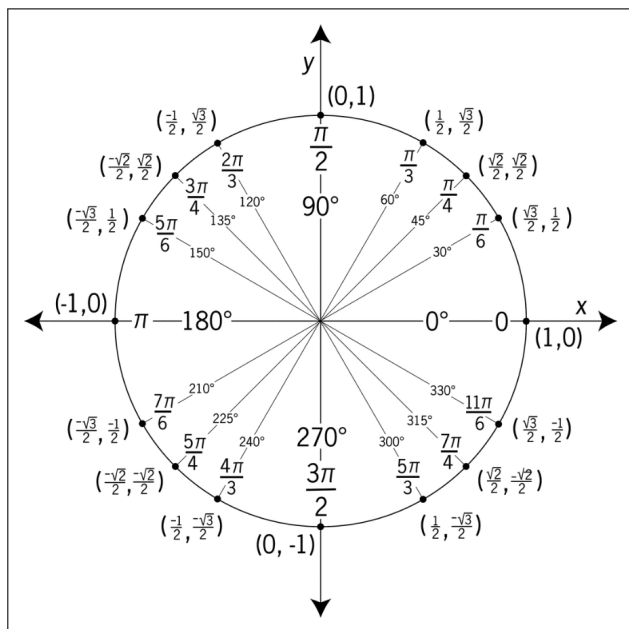
Cylindrical shells: $V = \int_a^b 2\pi x f(x) dx$

Polar coordinates

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

$$A = \int_a^b \frac{1}{2} (f(\theta)^2 - g(\theta)^2) d\theta$$

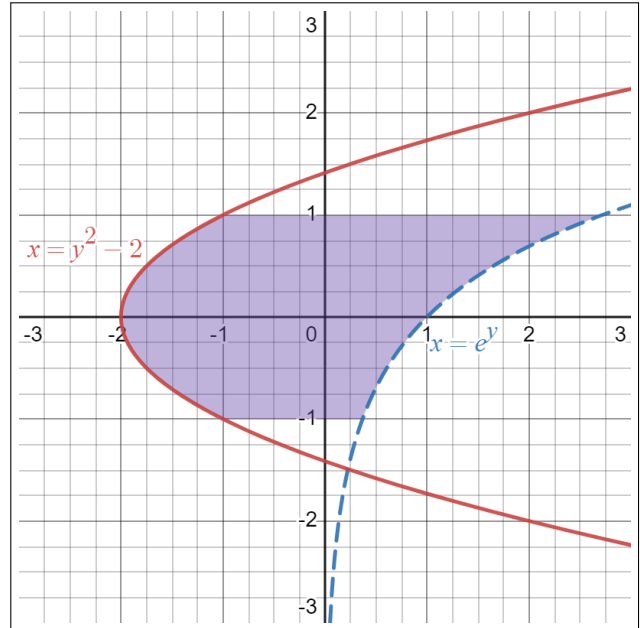
$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} d\theta$$



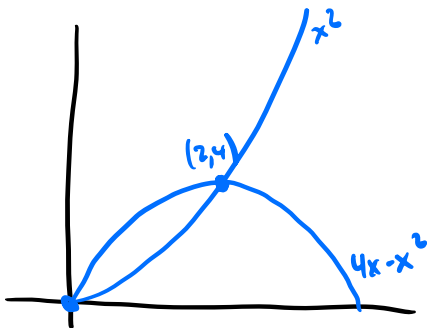
B1 - Areas between curves

- (a) Setup an integral representing the area of the shaded region shown to the right. You do not need to evaluate the integral. The curve graphed in a solid line is $x = y^2 - 2$, and the curve graphed in a dotted line is $x = e^y$.

$$\int_{-1}^1 (e^y - (y^2 - 2)) dy$$



- (b) Setup an integral representing the area enclosed by the curves $y = x^2$ and $y = 4x - x^2$. You do not need to evaluate the integral.

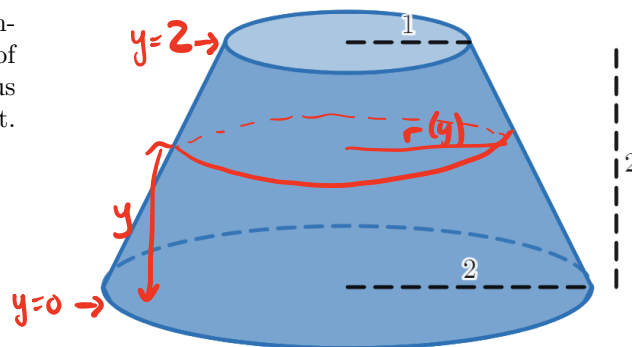


$$x^2 = 4x - x^2 \rightarrow 2x(2 - x) = 0 \rightarrow x = 0, 2$$

$$\int_0^2 (4x - x^2 - x^2) dx$$

B2 - Volumes

- (a) Use the method of cross sections to setup an integral representing the volume of a frustum of a right circular cone with height 2, lower radius 2, and upper radius 1, as shown on the right. You do not need to evaluate the integral.



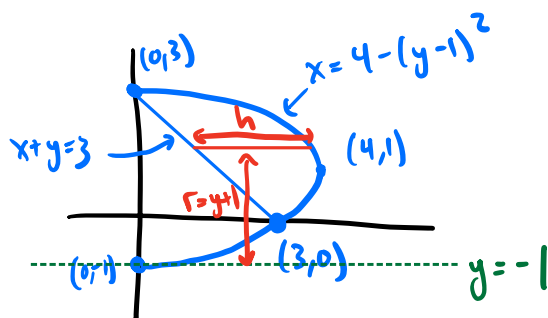
$r(y)$ is linear, with

$$r(0) = 2 \text{ and } r(2) = 1$$

$$r(y) = \frac{2-1}{0-2} (y-0) + 2 = -\frac{1}{2}y + 2$$

$$V = \int_0^2 \pi \left(-\frac{1}{2}y + 2\right)^2 dy$$

- (b) Setup an integral representing the volume of the solid obtained by rotating the region bounded by $x + y = 3$ and $x = 4 - (y - 1)^2$ about the line $y = -1$. You do not need to evaluate the integral.



horizontal slice parallel to axis of rotation $\rightarrow$ shells (along y -axis)

$$h = (4 - (y - 1)^2) - (3 - y)$$

$$V = \int_0^3 2\pi(y+1) \cdot ((4 - (y-1)^2) - (3-y)) dy$$

B3 - Arc length and surface area

- (a) (i) Find the arc length function of $f(t) = \frac{1}{27}(9t^2 + 6)^{\frac{3}{2}}$ starting at $x = 1$.

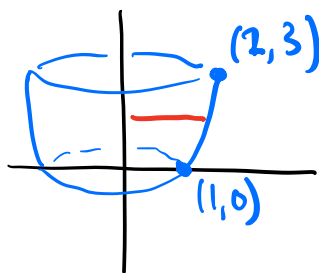
$$f'(t) = \frac{1}{18} \cdot (9t^2 + 6)^{\frac{1}{2}} \cdot 18t \rightarrow 1 + f'(t)^2 = 1 + t^2(9t^2 + 6) = 9t^4 + 6t^2 + 1 = (3t^2 + 1)^2$$

$$s(x) = \int_1^x \sqrt{(3t^2 + 1)^2} dt = \int_1^x (3t^2 + 1) dt = t^3 + t \Big|_1^x = \boxed{x^3 + x - 2}$$

- (ii) What is the arc length of $f(t)$ from $t = 1$ to $t = 2$?

$$s(2) = 2^3 + 2 - 2 = \boxed{8}$$

- (b) Setup two different integrals representing the surface area of the shape obtained by rotating the curve $y = x^2 - 1$, $0 \leq y \leq 3$, about the y -axis. One of your integrals should involve integrating along the x -axis and one should involve integrating along the y -axis. You do not need to evaluate either integral.



$$r = x = \sqrt{1 + y} = g(y)$$

$$f'(x)^2 + 1 = (2x)^2 + 1$$

$$g'(y)^2 + 1 = \frac{1}{4(1+y)} + 1$$

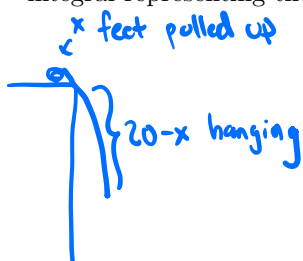
$$S = \int_1^2 2\pi x \sqrt{1 + 4x^2} dx$$

or

$$S = \int_0^3 2\pi \sqrt{1+y} \sqrt{\frac{1}{4(1+y)} + 1} dy$$

B4 - Applications of integration to physics and engineering

- (a) A chain is 20 feet long and weighs 60 pounds. It hangs from a building, and a ribbon is tied around the midpoint of the chain (i.e. 10 feet below the top of the building). You sit at the top of the building and pull the chain up until the ribbon reaches the top of the building. Setup, but do not evaluate, an integral representing the total work that you did.



force when x feet have been pulled up:

$$F(x) = \frac{60}{20} \frac{\text{lbs}}{\text{ft}} \cdot (20-x) \text{ft} = 3 \cdot (20-x) \text{ lbs.}$$

$$W = \int_0^{10} 3 \cdot (20-x) dx$$

- (b) Setup, but do not evaluate, an integral representing the total hydrostatic force felt by the trapezoid submerged under water as shown on the right. Use ρ for the density of the water and g as the acceleration of gravity.

$$w(1) = 4$$

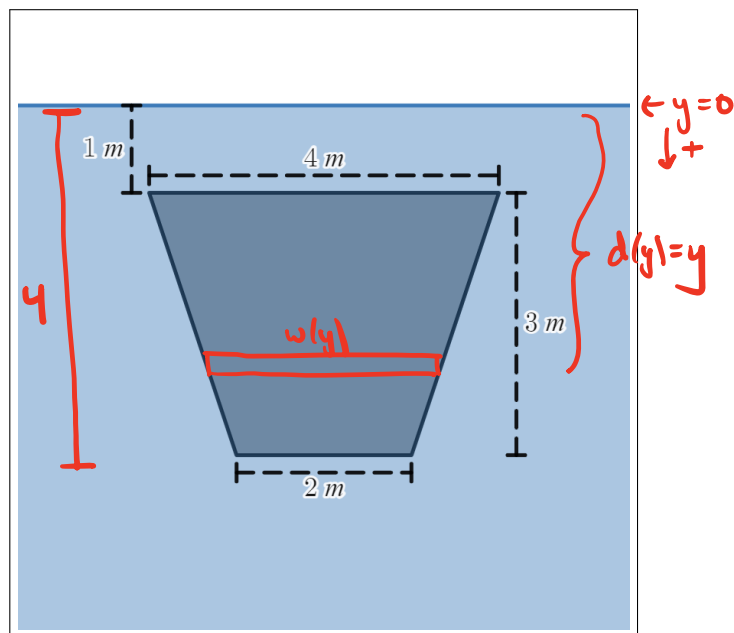
$$w(4) = 2$$

$$w(y) = \frac{4-2}{1-4} (y-1) + 4$$

$$= -\frac{2}{3} (y-1) + 4$$

$$F = \int_1^4 \rho g \cdot d(y) \cdot w(y) dy$$

$$= \int_1^4 \rho g \cdot y \cdot \left(-\frac{2}{3}(y-1) + 4\right) dy$$



B5 - Areas and lengths in polar coordinates

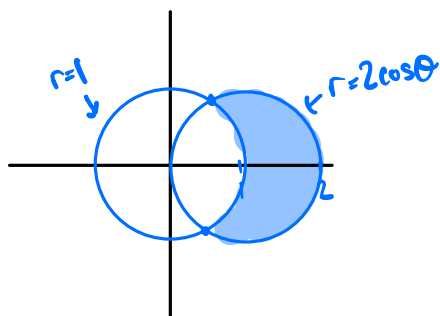
- (a) Find the values of θ where the curve $r = 3 \cos \theta$, $0 \leq \theta < \pi$, has a horizontal or vertical tangent line. For full credit, you must use calculus to answer this question.

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} = \frac{-3 \sin^2 \theta + 3 \cos^2 \theta}{-3 \sin \theta \cos \theta - 3 \cos \theta \sin \theta} = \frac{\cos(2\theta)}{-\sin(2\theta)}$$

$$\text{horizontal: } \cos(2\theta) = 0 \rightarrow \boxed{\theta = \frac{\pi}{4}, \frac{3\pi}{4}}$$

$$\text{vertical: } \sin(2\theta) = 0 \rightarrow \boxed{\theta = 0, \frac{\pi}{2}}$$

- (b) Setup an integral representing the area of the region that lies inside of $r = 2 \cos \theta$ and outside of $r = 1$. You do not need to evaluate the integral.



$$2 \cos \theta = 1 \rightarrow \theta = -\frac{\pi}{3}, \frac{\pi}{3}$$

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{1}{2} ((2 \cos \theta)^2 - 1^2) d\theta$$